Figure 9.11 Initiation of translation in eukaryotic cells

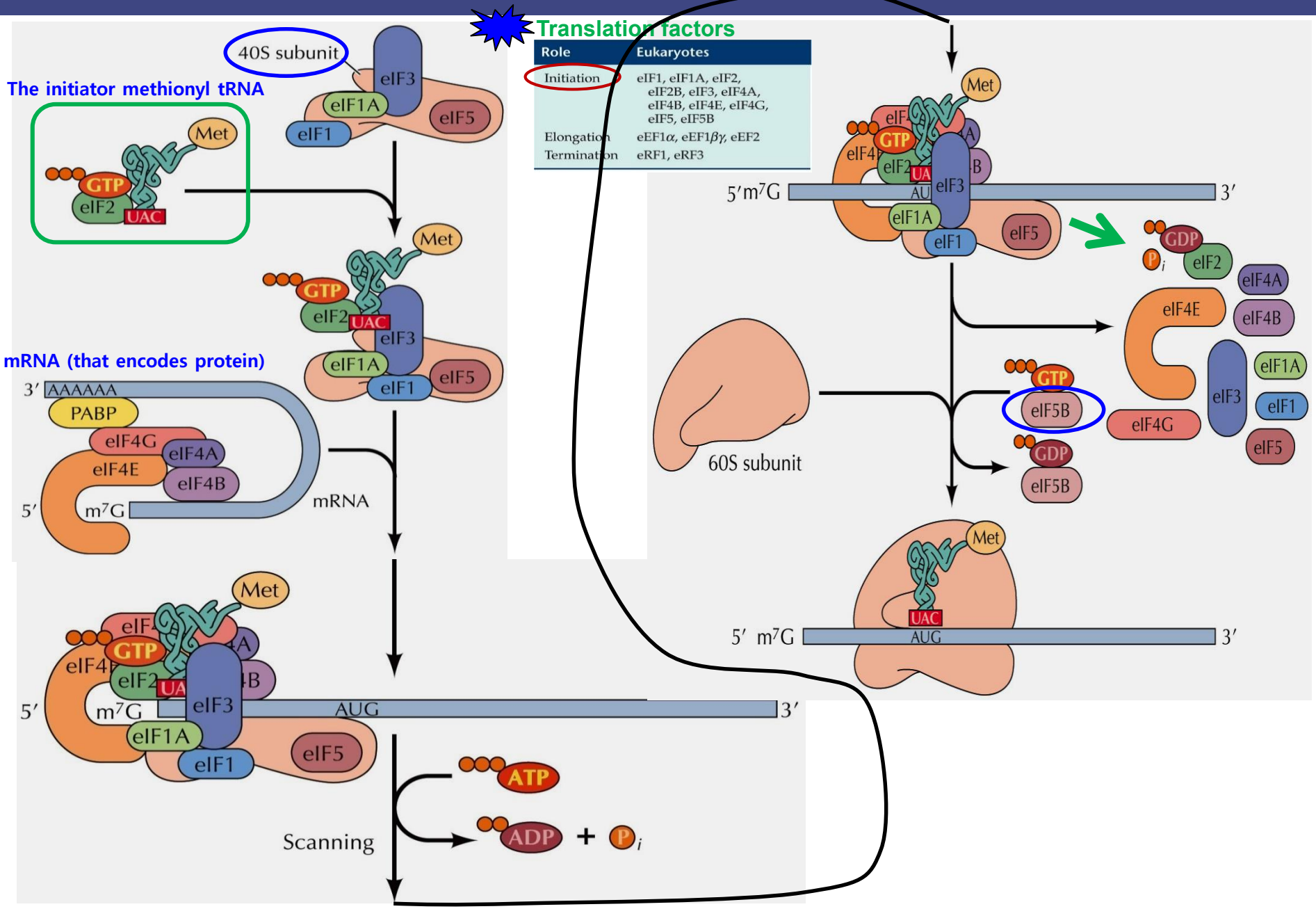
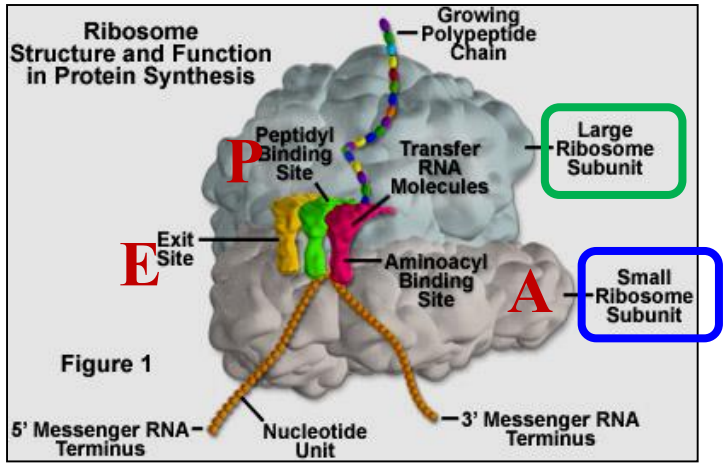


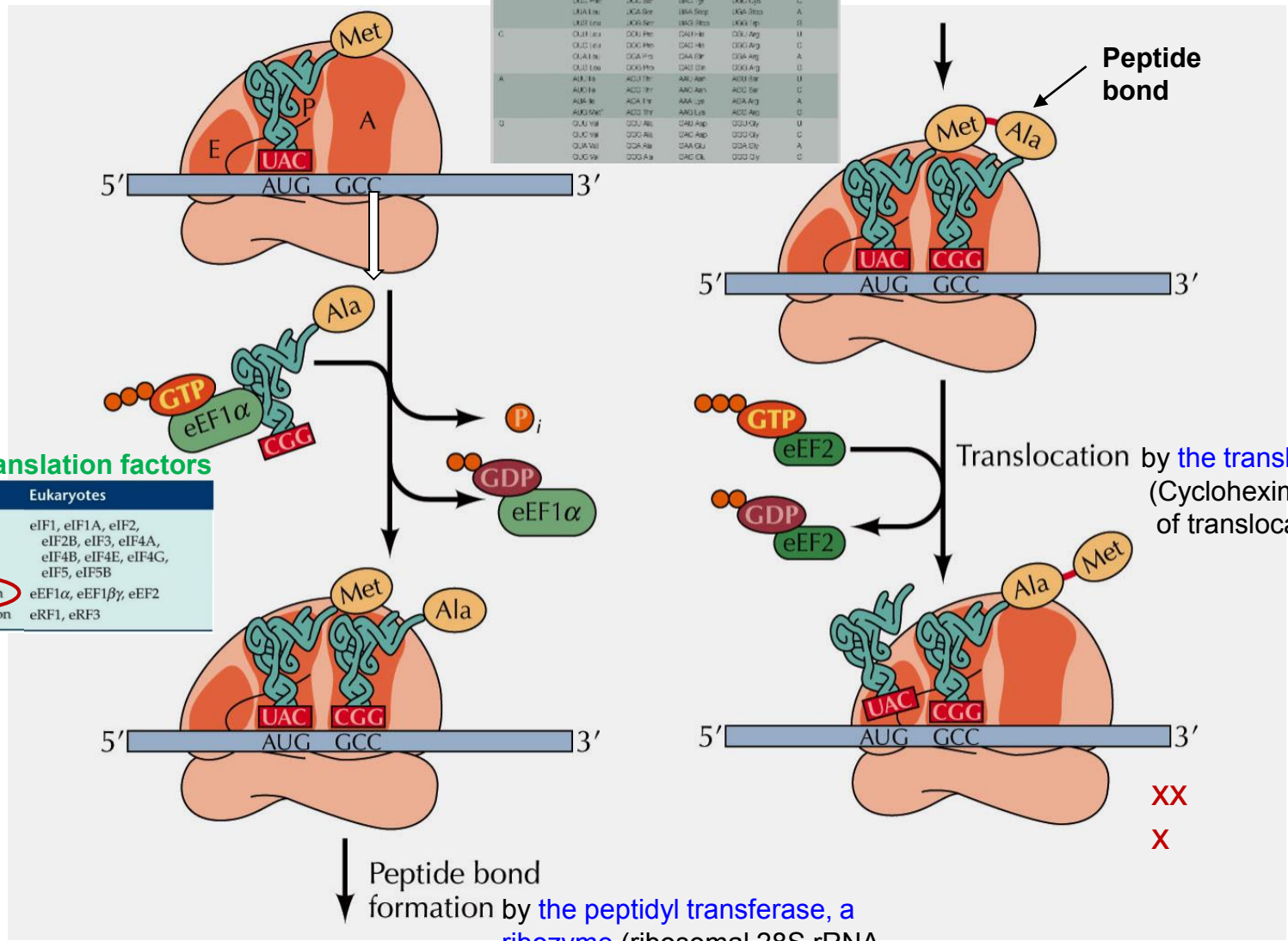
Figure 9.13 Elongation stage of translation



▶ **The large ribosomal subunit** has **three tRNA binding sites**: **P (peptidyl), A (aminoacyl), and E (exit) sites**.

Translation factors

Role	Eukaryotes
Initiation	eIF1, eIF1A, eIF2, eIF2B, eIF3, eIF4A, eIF4B, eIF4E, eIF4G, eIF5, eIF5B
Elongation	eEF1α, eEF1$\beta\gamma$, eEF2
Termination	eRF1, eRF3



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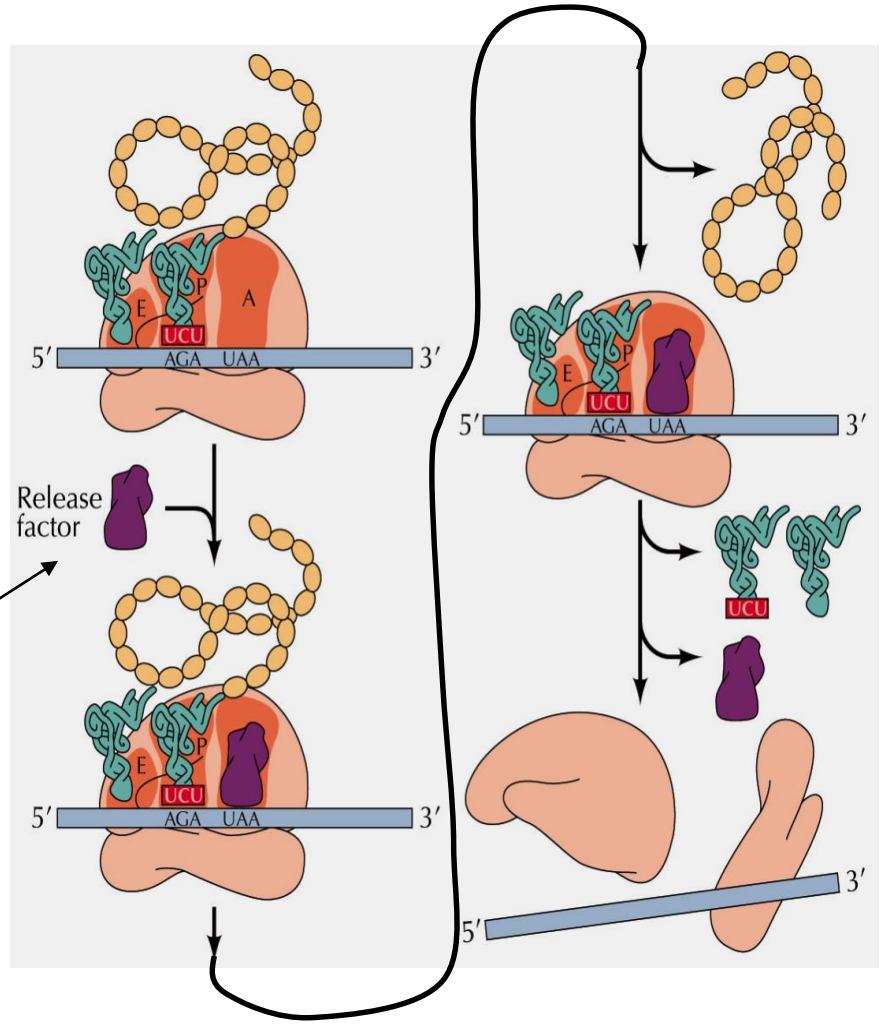
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Figure 9.15 Termination of translation

- **Elongation continues** until **a stop codon (UAA, UAG or UGA)** is translocated into **the A site**.
- **Release factors (RF)** recognize the signals and **terminate protein synthesis**.
- In eukaryotic cells, **eRF1 recognizes all three stop codons.**

Translation factors

Role	Eukaryotes
Initiation	eIF1, eIF1A, eIF2, eIF2B, eIF3, eIF4A, eIF4B, eIF4E, eIF4G, eIF5, eIF5B
Elongation	eEF1α, eEF1βγ, eEF2
Termination	eRF1, eRF3



Translation of mRNA

- **A single mRNA can be translated simultaneously by a series of multiple ribosomes in both pro- and eukaryotic cells**: Once a ribosome has moved away from the initiation site of mRNA, another ribosome can bind to the mRNA and begin synthesis of a new polypeptide chain.
- **Thus mRNAs are usually translated by a series of ribosomes, spaced at intervals of 100 to 200 nucleotides.**
- **The group of ribosomes bound to an mRNA molecule** is called **a polyribosome** (also called **polysome**).
- **Each ribosome within polysome functions independently to synthesize a separate polypeptide chain**.

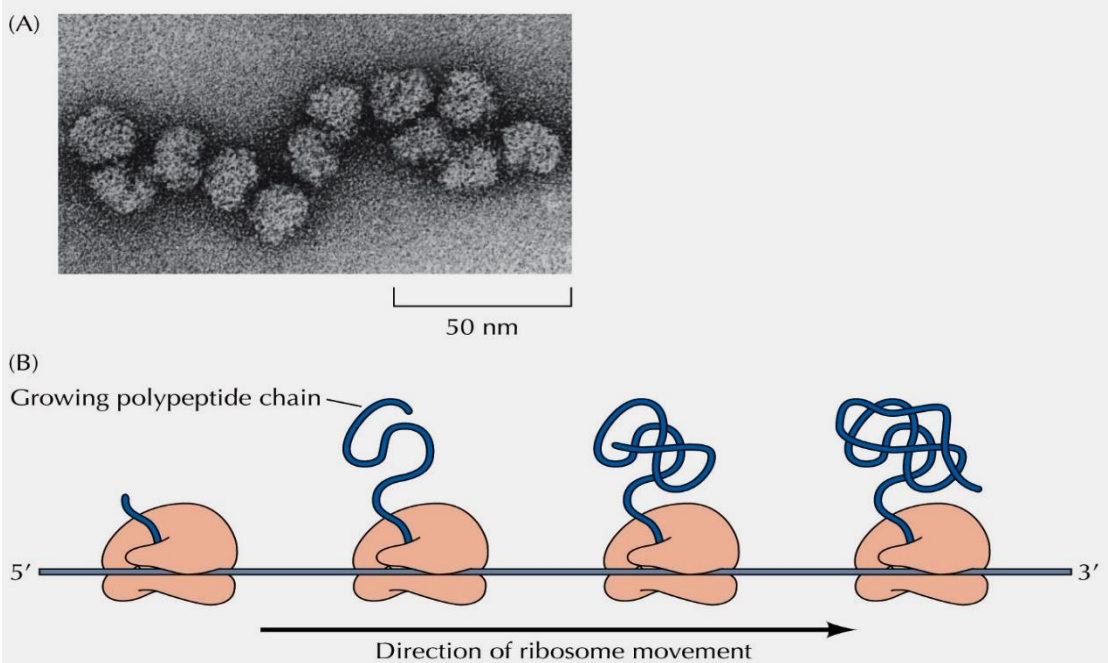


Figure 9.16. (A) EM of a eukaryotic polysome and (B) Schematic of a generalized polysome

Lecture Contents

1. Introduction about learning performance of today's lecture
2. Translation (protein synthesis) of mRNA in eukaryotes
   - Overview of translation, codon, tRNA, ribosome
3. Three steps of translation in eukaryotes
   - Translation initiation, elongation, termination
   - Translation factors involved in each step
4. **Regulation of translation in eukaryotes**

Regulation of translation

- **Regulation of translation** also plays a key role in modulating gene expression.
- Regulation of translation includes **translational repressor proteins**, **microRNAs**, and **many other cellular factors**.
- **Global translational activity of cells is modulated in response to cell stress, nutrient availability, and growth factor stimulation**.

▶ **Translation of a ferritin mRNA** encoding **a ferritin protein that stores iron in cells** is regulated by **repressor proteins**, **depending on cellular levels of iron**.

▶ **When iron is absent**, **iron regulatory protein (IRP)** in cells binds to **the iron response element (IRE) in the 5′ UTR**, thereby **blocking translation**.

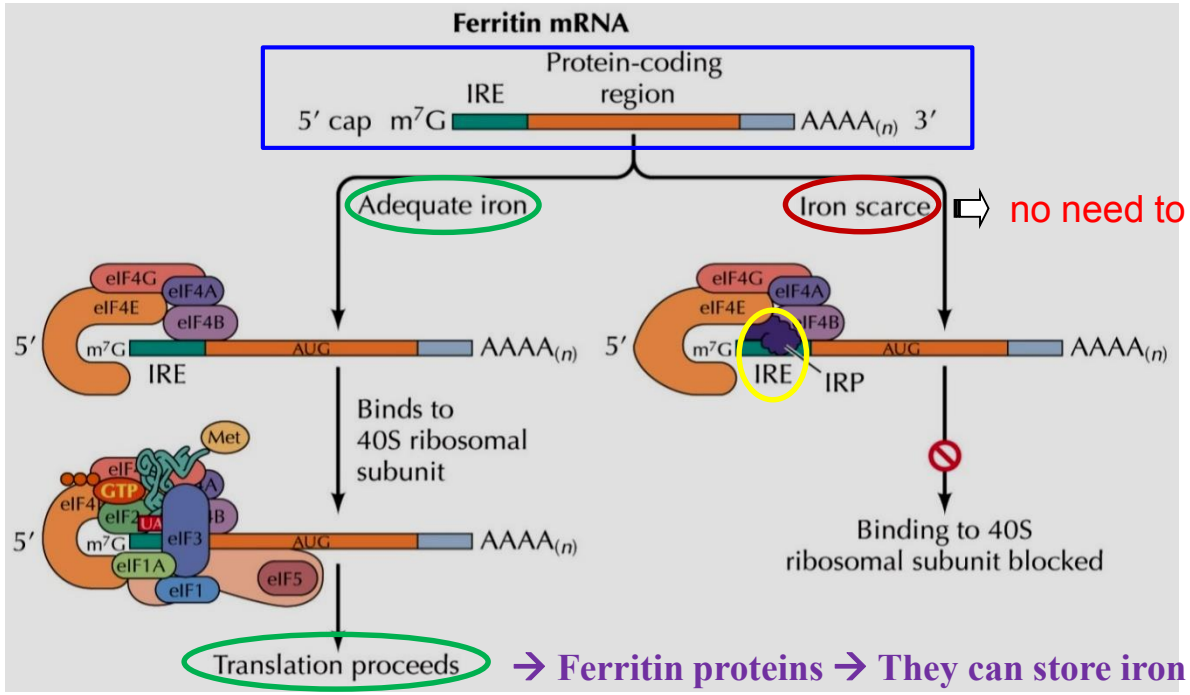


Figure 9.17. Translational regulation of ferritin

Lecture Contents

1. Introduction about learning performance of today's lecture
2. Translation (protein synthesis) of mRNA in eukaryotes
   - Overview of translation, codon, tRNA, ribosome
3. Three steps of translation in eukaryotes
   - Translation initiation, elongation, termination
   - Translation factors involved in each step
4. Regulation of translation in eukaryotes

2026-2학기 세포의 분자적 이해

9/28 (월)

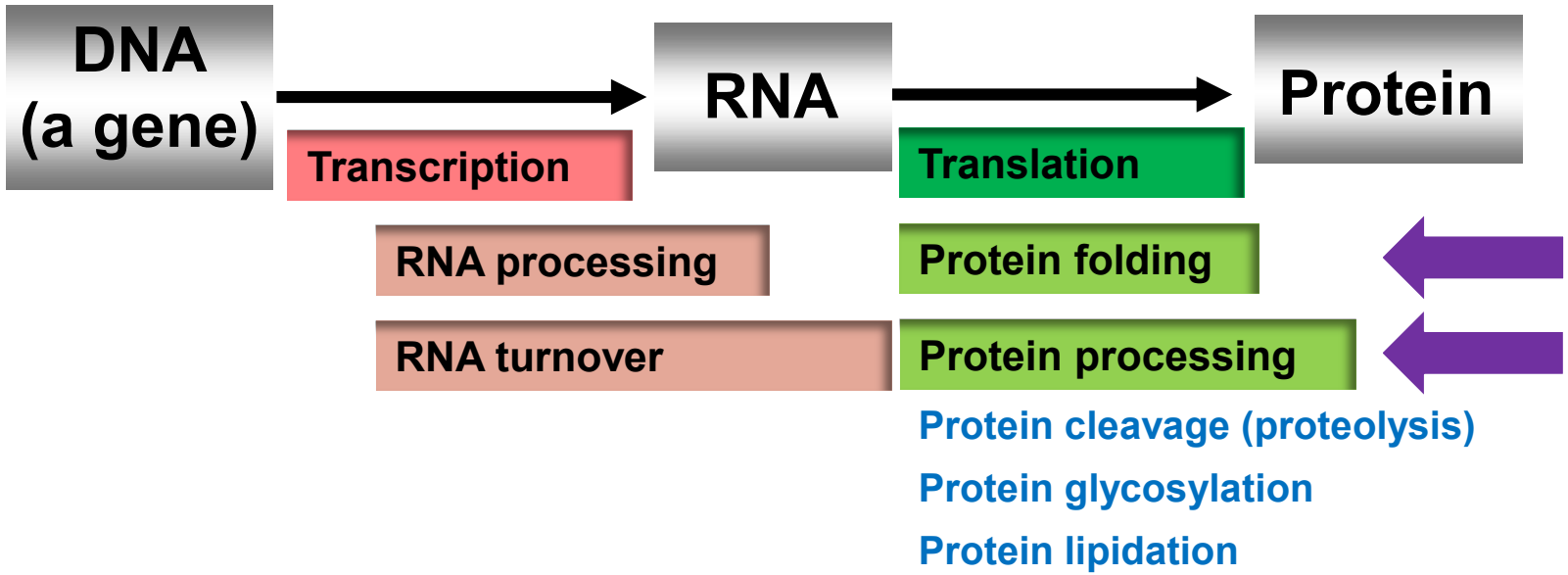
Chapter 9.
Protein Synthesis, Processing, and Regulation (2)

분자의학교실 장병철

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Lecture Contents

1. Introduction about learning performance of today's lecture
2. Protein folding in eukaryotes
3. Protein processing in eukaryotes



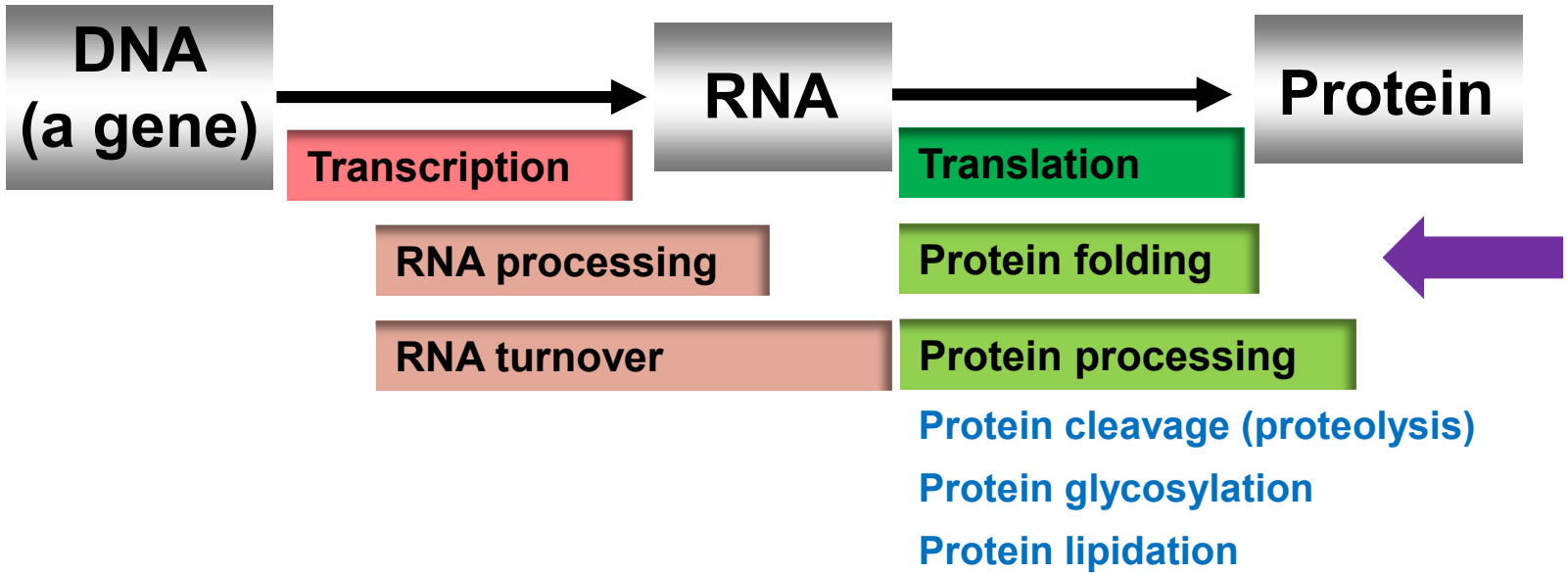
Learning Performance of today's lecture

1) Students can explain the protein folding in eukaryotic cells.

2) Students can explain the types of protein processing in eukaryotic cells.

Lecture Contents

1. Introduction about learning performance of today's lecture
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2. Protein folding in eukaryotes

► Protein folding (and processing) in eukaryotes

- **Newly synthesizing proteins** (polypeptide chains) must undergo **folding** (and **processing**) to become **functional proteins**.

► Protein folding and Chaperone Proteins in eukaryotes

- All newly-made proteins require assistance to **convert from a linear chain of amino acids to a functional three-dimensional (3-D) entity.**
- Requires the chemical energy ATP.
- Mediated by **chaperone proteins**, including **cytosolic chaperones, mitochondrial chaperones, heat shock protein (hsp), and others**, that facilitate the folding of proteins.
- For that, chaperone proteins bind to nascent polypeptide chains that are still being translated on ribosomes.
- The nascent polypeptide chains **must be protected from aberrant folding (aggregation)** with other proteins until synthesis of an entire domain is complete.

	Chaperone proteins	
Protein family	**Prokaryotes**	**Eukaryotes**
Hsp70	DnaK	Hsc73 (cytosol) BiP (endoplasmic reticulum) SSC1 (mitochondria) ctHsp70 (chloroplasts)
Hsp60	GroEL	TriC (cytosol) Hsp60 (mitochondria) Cpn60 (chloroplasts)
Hsp90	HtpG	Hsp90 (cytosol) Grp94 (endoplasmic reticulum)

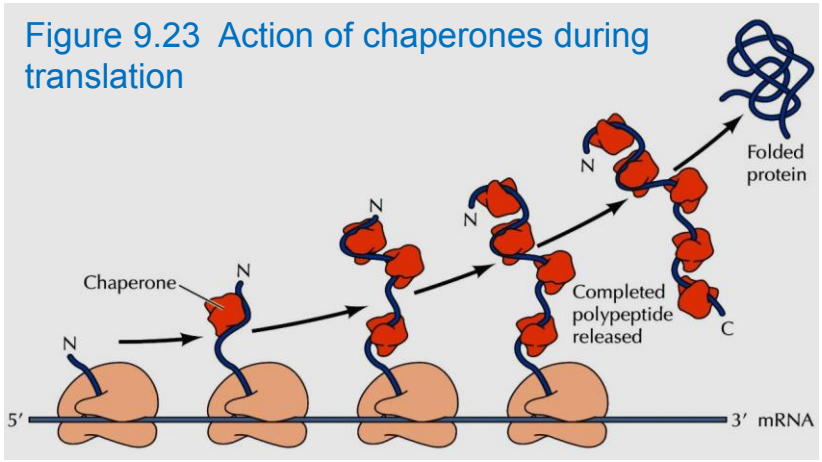
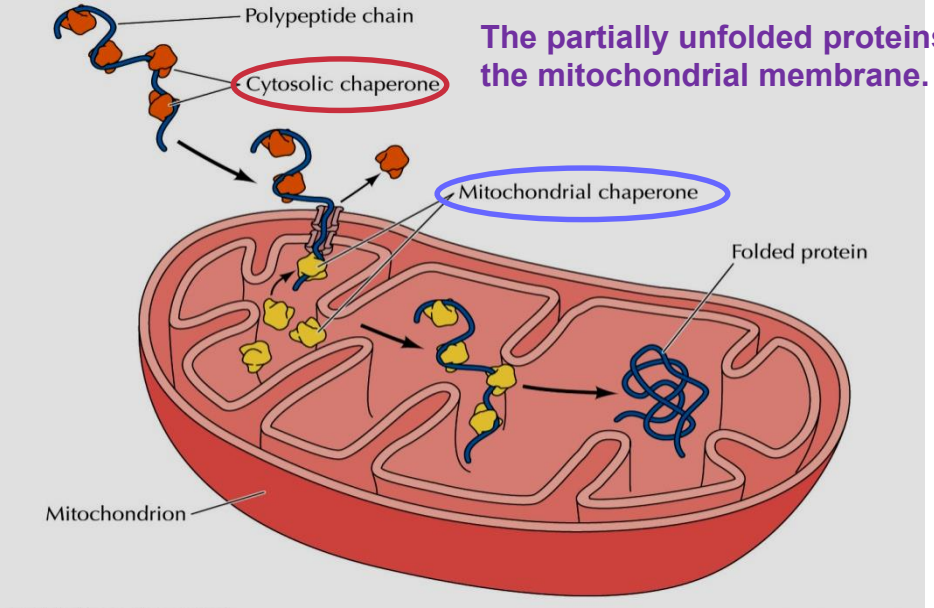


Figure 9.23 Action of chaperones during translation

2. Protein folding in eukaryotes

► Actions of chaperones (cytosolic, mitochondrial) during protein transport

- They stabilize unfolded polypeptide chains during their transport into organelles.
- Mitochondrial chaperones in the mitochondrion then facilitate subsequent folding.



The partially unfolded proteins stabilized by cytosolic chaperones are transported across the mitochondrial membrane.

Figure 9.24 Action of chaperones during protein transport

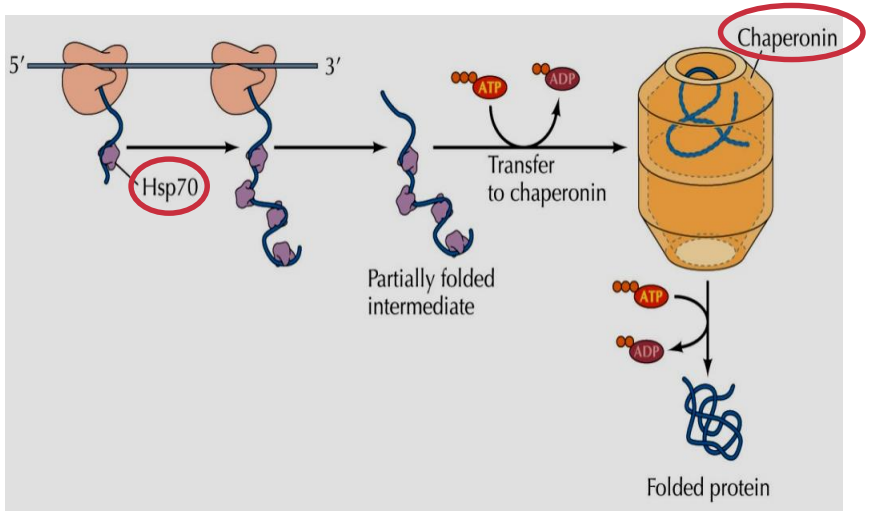


Figure 9.25 Sequential actions of chaperones

► Hsp70 chaperones and chaperonins

- Hsp70 chaperones and chaperonins are found in both eukaryotic and prokaryotic cells.
- **Hsp70 proteins** stabilize unfolded polypeptide chains during translation by binding to short hydrophobic segments.
- The unfolded polypeptide is then transferred to **a chaperonin**, **a form of a double-chambered structure where the unfolded protein folding takes place**.

2. Protein folding in eukaryotes

► **Enzymes that function as chaperones** by facilitating protein folding

(1) Protein disulfide isomerase (PDI)

- PDI **catalyzes disulfide bond formation**.
- PDI is one of the most abundant proteins **in the ER**, **where an oxidizing environment allows (S-S) linkages**.

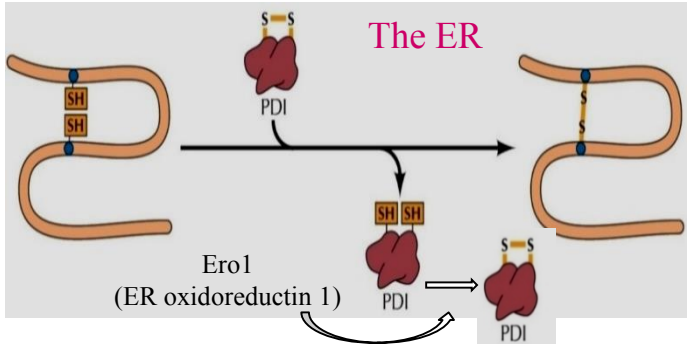


Figure 9.28 The action of PDI

(2) Peptidyl prolyl isomerase (PPI)

- The PPI family is **a group of enzymes** that **catalyze the cis ↔ trans isomerization of proline-containing peptide bonds**, which is crucial for **protein folding, signaling, and regulation**.

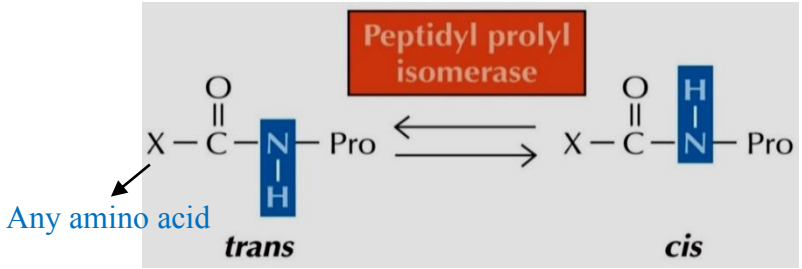


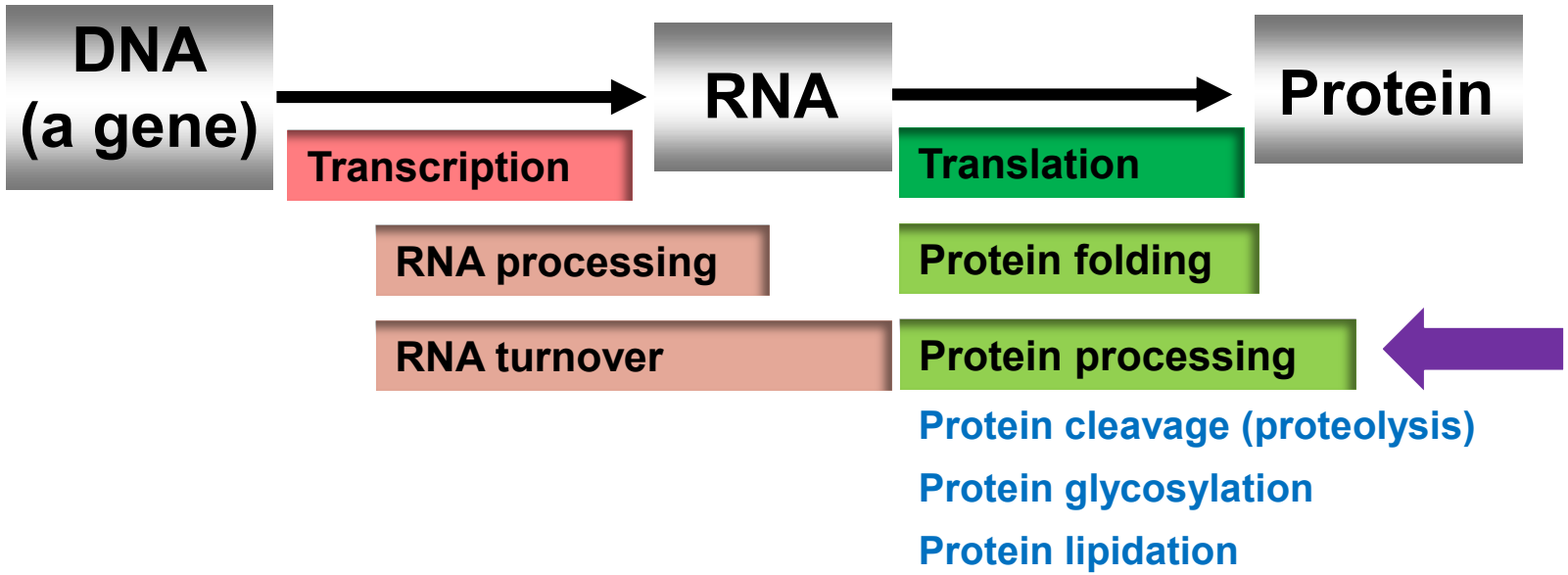
Figure 9.28a The action of PPI

PPI

- **In general**, **peptide bonds between amino acids** are **more stable** in the **trans** conformation.
- However, because of **its unique ring structure**, **proline** can also exist in the **cis** conformation **with relatively high stability**. **The problem is that during protein folding, the cis ↔ trans isomerization of proline-containing peptide bonds occurs very slowly.**
- **PPI catalyze this conversion, allowing it to occur much more rapidly**.

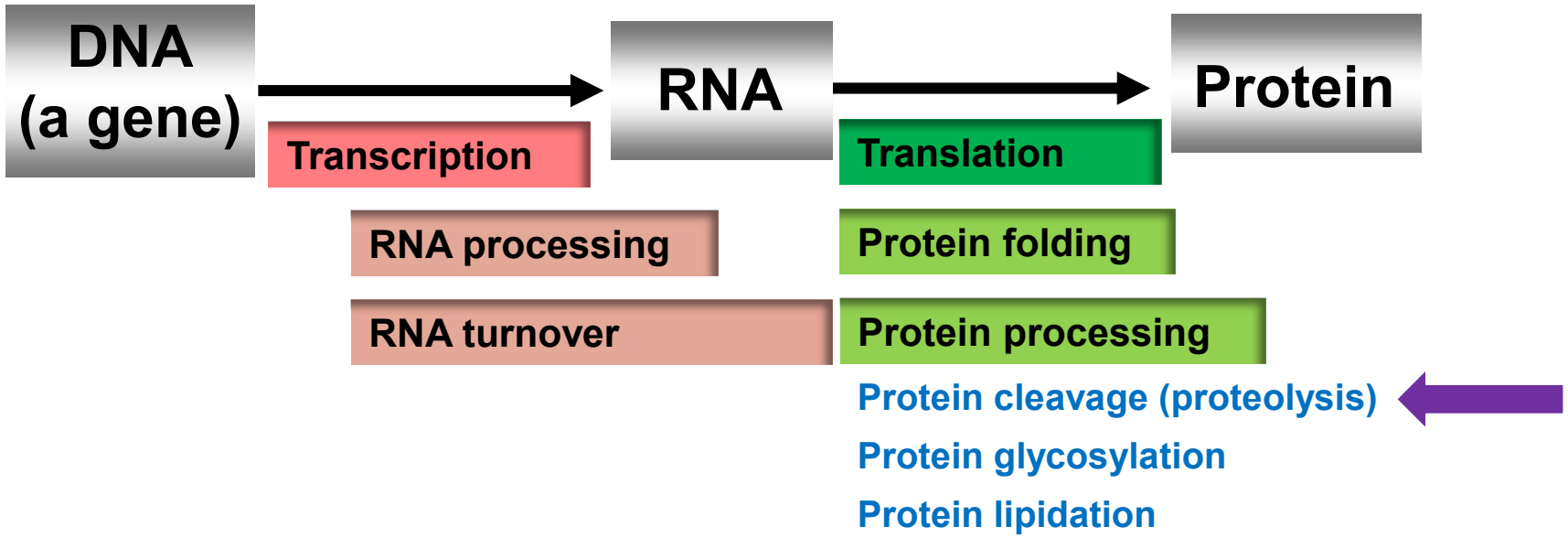
Lecture Contents

1. Introduction about learning performance of today's lecture
2. Protein folding in eukaryotes
3. **Protein processing in eukaryotes**



Lecture Contents

1. Introduction about learning performance of today's lecture
2. Protein folding in eukaryotes
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3. Protein processing in eukaryotes: Protein cleavage (proteolysis)

► Protein cleavage (proteolysis)

: **Partial (or total) cleavage** of the polypeptide chain is **an important step in the maturation of many proteins.**

(1) Partial proteolysis: ER proteins

- Some proteins have **amino (N)-terminal signal sequences** that target the protein for transport to a specific destination like **the endoplasmic reticulum (ER).**
- Signal peptide is removed by **signal peptidase** in the ER.

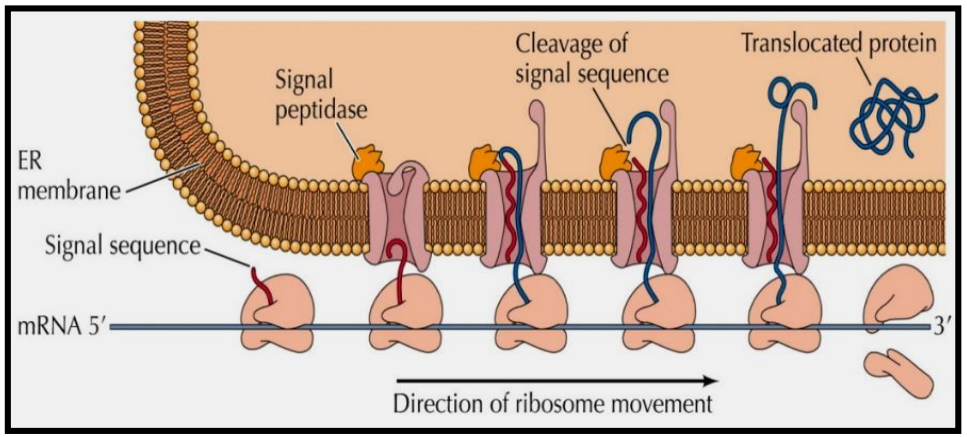


Figure 9.30 The role of signal sequences of the newly synthesized protein in membrane translocation

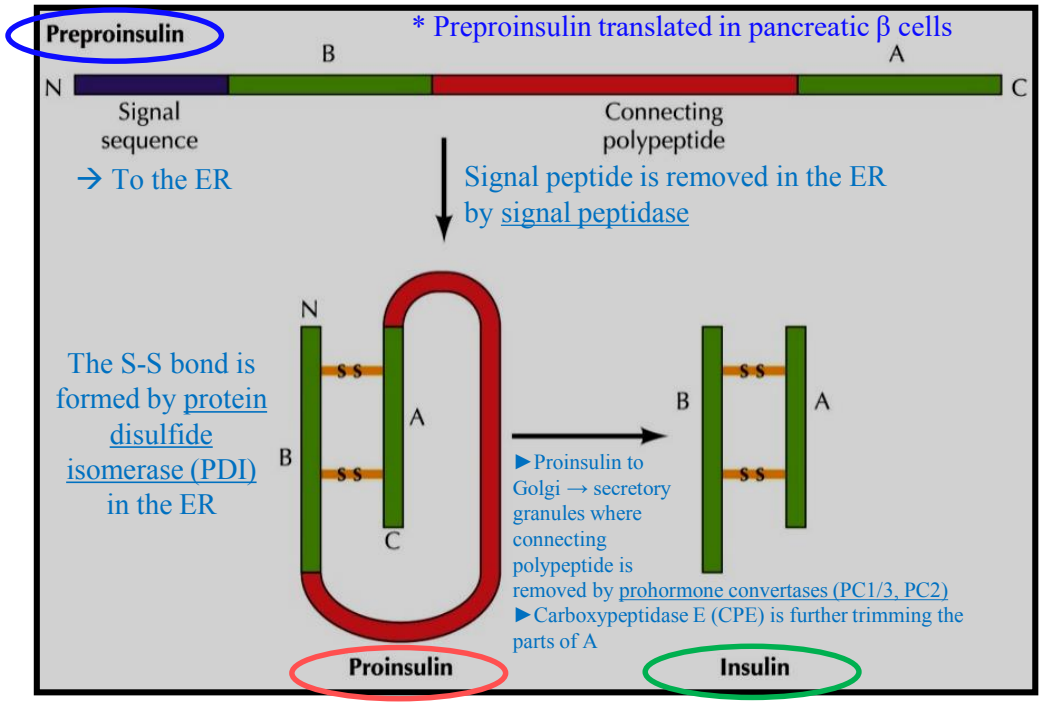
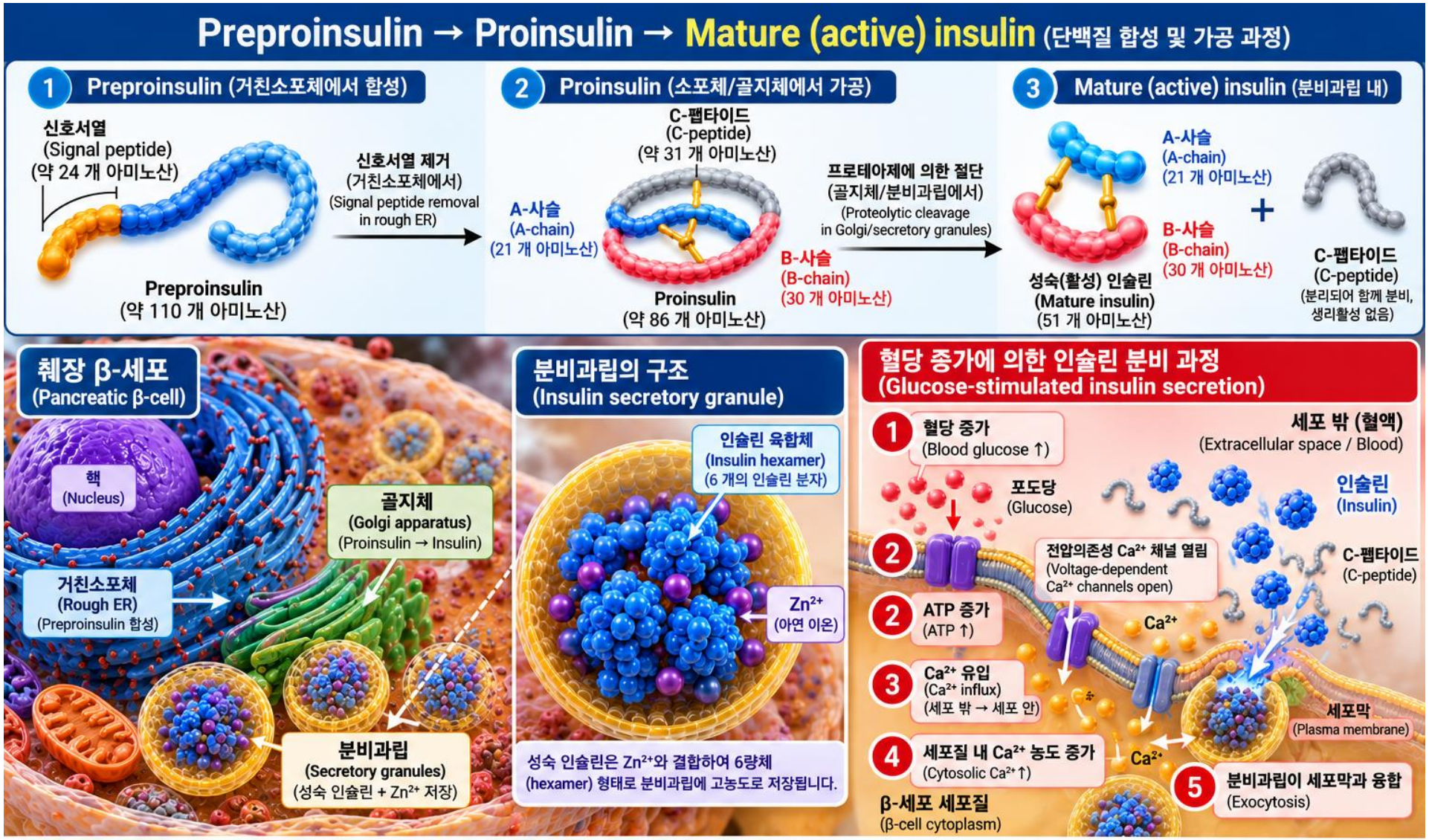


Figure 9.31 Proteolytic processing of insulin

(2) Partial proteolysis: Insulin

- Proteolytic processing also **includes formation of mature hormones (insulin, etc) or active enzymes (caspases, etc) by partial cleavage of larger precursors**.
- **The hormone insulin** is synthesized as a precursor polypeptide that **goes through 2 cleavages to produce the mature insulin**.

3. Protein processing in eukaryotes: Protein cleavage (proteolysis)

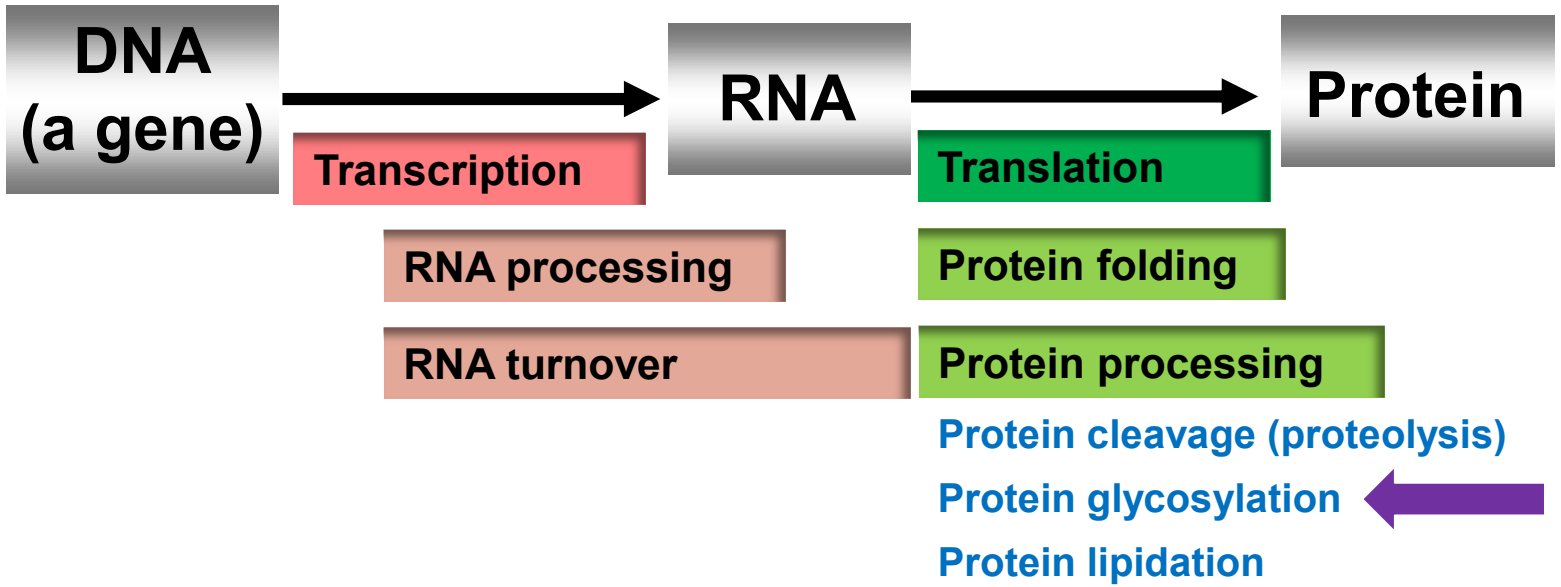


핵심 정리

- 인슐린은 거친소포체에서 preproinsulin으로 합성되고, 가공을 거쳐 성숙 인슐린이 됩니다.
- 성숙 인슐린은 Zn^{2+}와 함께 6량체 형태로 분비과립에 저장됩니다.
- 혈당이 증가하면 Ca^{2+}가 세포 밖에서 β-세포 안으로 유입되어 분비과립이 세포막과 융합하고, 인슐린이 세포 밖(혈액)으로 분비됩니다.
- C-펩타이드는 인슐린과 함께 분비되지만 생리활성이 없습니다.

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1. Introduction about learning performance of today's lecture
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3. Protein processing in eukaryotes: Glycosylation

► **Glycosylation : Addition of carbohydrate chains to proteins → glycoproteins**

(1) **The carbohydrate chains (moieties)** play important roles in protein folding in the ER, in **targeting proteins for transport**, **for activity**, and as **recognition sites in cell-cell interactions**.

(2) **N-linked glycoproteins**: the carbohydrate is attached to **the N atom in the side chain of asparagine (Asn, N)**.

(3) **O-linked glycoproteins**: the carbohydrate is attached to **the O atom in the side chain of serine (Ser, S) or threonine (The, T)**.

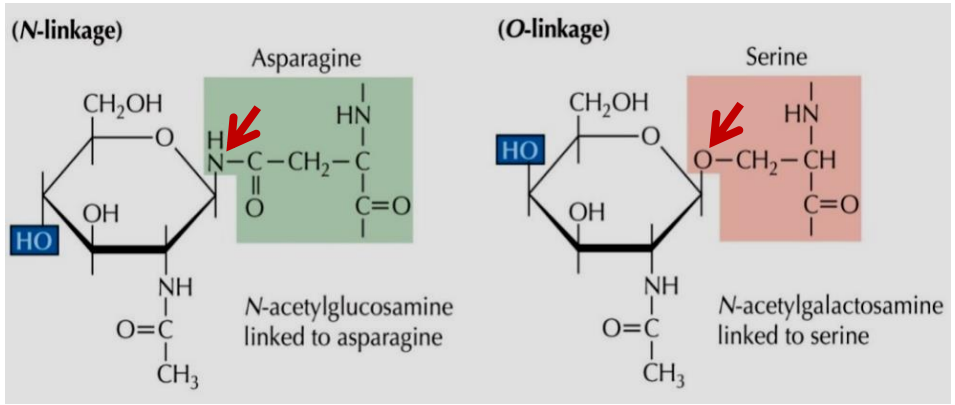


Figure 9.32 Linkage of carbohydrate side chain to glycoproteins

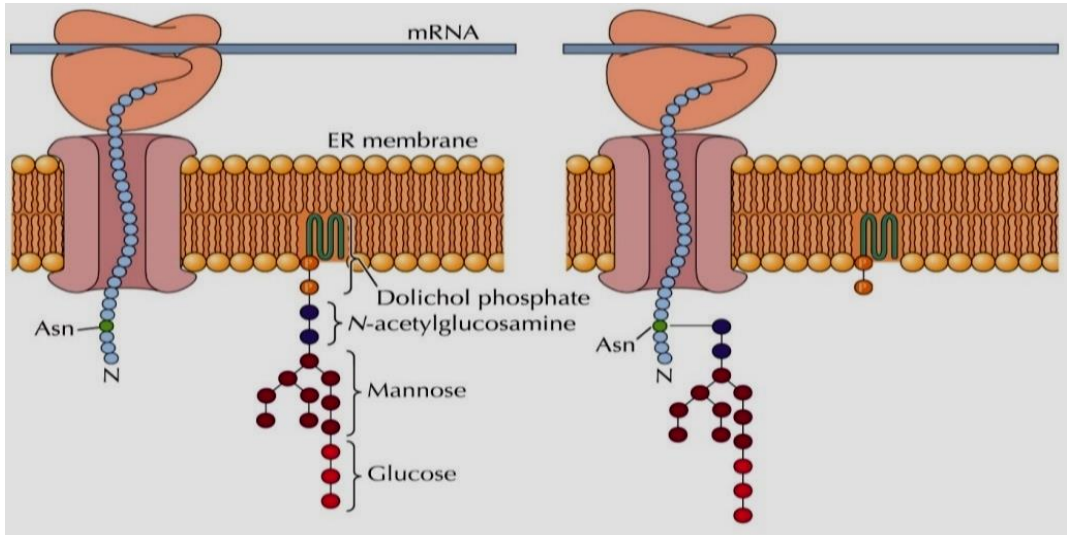


Figure 9.33 Synthesis of N-linked glycoproteins (eg: COX-2, the PGs producing enzyme from AA)

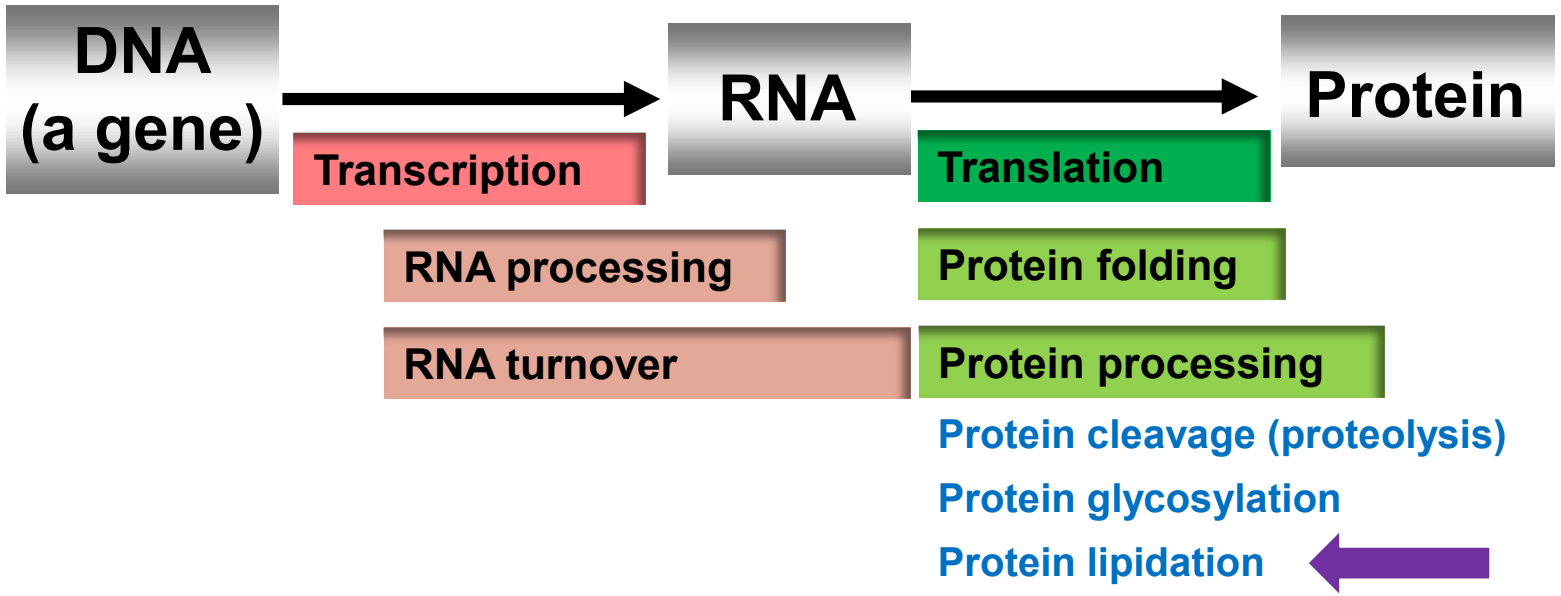
(4) **A 14-sugar oligosaccharide** is transferred to an Asn residue of the growing polypeptide chain.

(5) The oligosaccharide is assembled on **a lipid carrier (dolichol phosphate) in the ER membrane**.

The N-glycosylation of protein (COX-2, etc) starts in the ER before translation is complete.

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3. Protein processing in eukaryotes: Lipidation

► Lipidation : Attachment of lipids (fatty acids, FAs) to proteins (lipoproteins)

- Some eukaryotic proteins are modified with lipids (FAs), which often **serve to anchor them to the plasma membrane**.
- There are four types of lipid additions: N-myristoylation, palmitoylation, prenylation, glycolipids

(1) N-myristoylation

- **Myristic acid (a 14-carbon FA)** is attached to **an N-terminal glycine**.
- These proteins including **Src** are **associated with the inner face of the plasma membrane**.

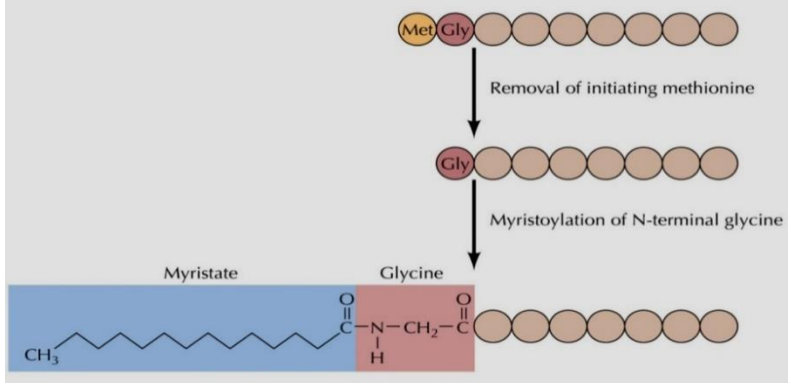
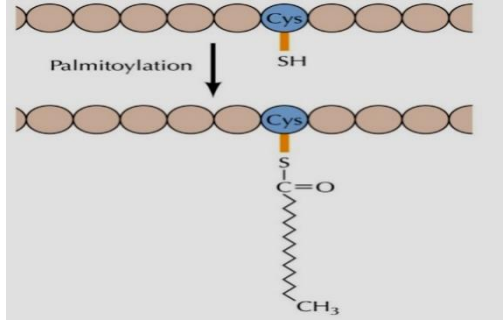


Figure 9.35 Addition of myristic acid by N-myristoylation

(2) Palmitoylation

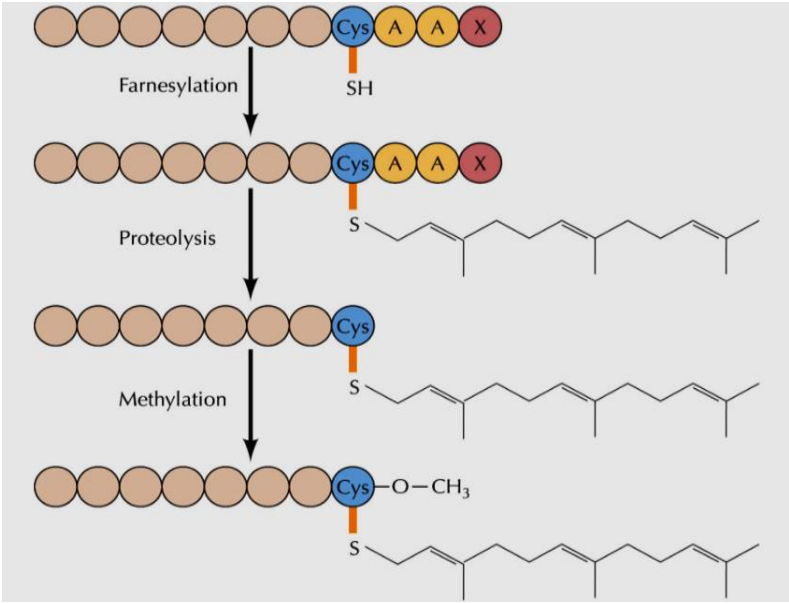
- **Palmitic acid (a 16-carbon FA)** is added to **sulfur atoms of the side chains of internal cysteine residues**.

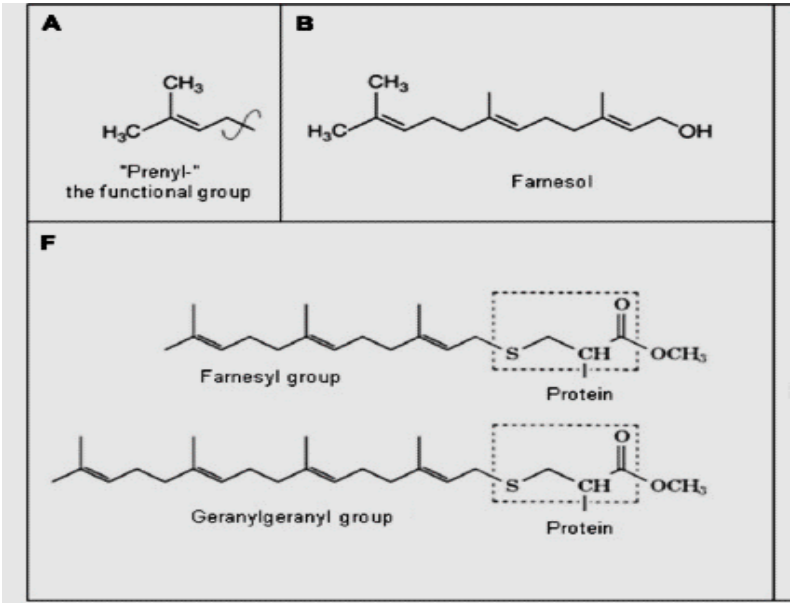


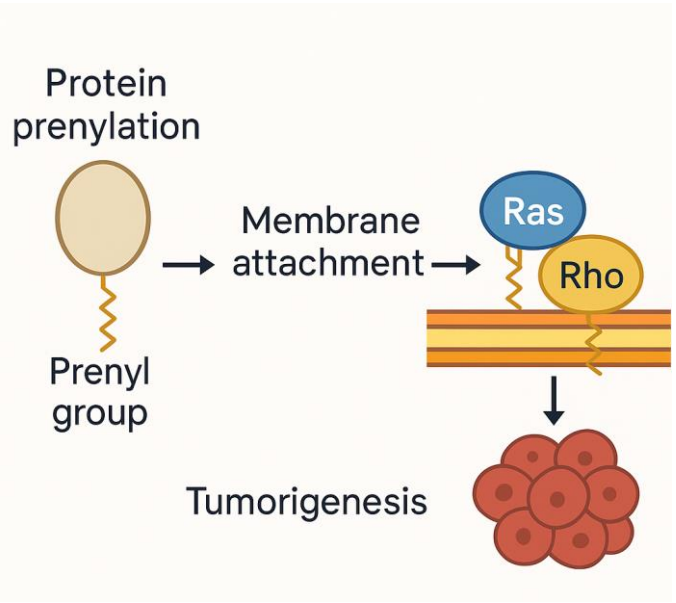
3. Protein processing in eukaryotes: Lipidation

(3) Prenylation

- **Protein prenylation** (also known as isoprenylation) is **the addition of hydrophobic molecules to a protein** (or chemical compound).
- This process involves **the transfer of prenyl groups (either a farnesyl or a geranylgeranyl moiety) to sulfur atoms** in the side chains of **cysteine(s) of C-terminal of the target protein**.
- **Three enzymes** that carry out prenylation in the cell include **farnesyl transferase**, **Caax protease** and **geranylgeranyl transferase I**.
- **Farnesyl transferase inhibitors (FTIs)** inhibit this process.
- It is an important process to mediate **protein–membrane interactions and protein–protein interactions (PPIs).**
- Many of these proteins are involved in **control of cell growth and differentiation**, including **Ras oncoprotein (and its abnormal prenylation)**, which is responsible for **many human cancers**. FTIs are considered to be anti-cancer drugs.





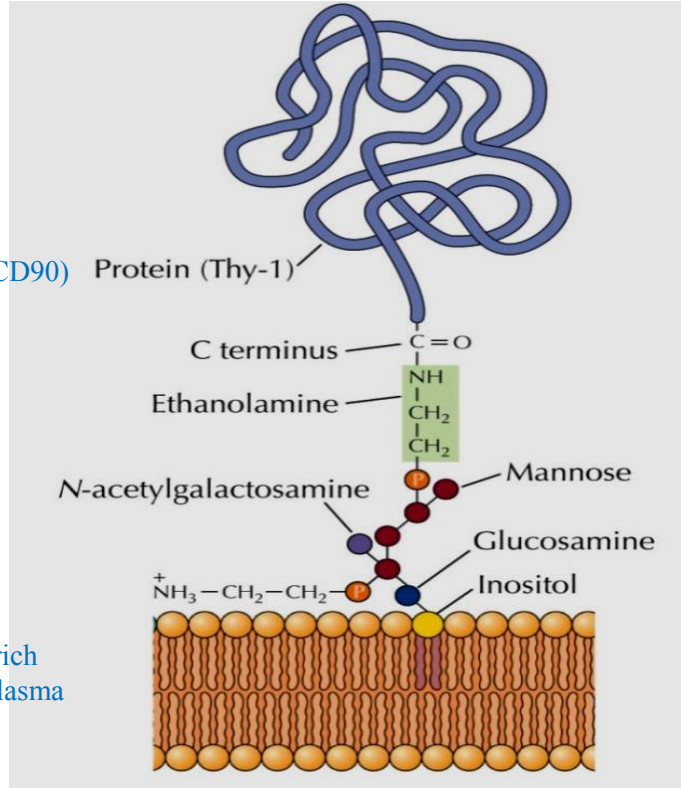


3. Protein processing in eukaryotes: Lipidation

(4) GPI (glycosylphosphatidylinositol) anchoring (also called Glycolipids)

- GPI anchoring is a type of protein lipidation that occurs during post-translational processing.
- **A GPI anchor** is attached to the C-terminus of certain proteins, anchoring them to the outer leaflet of the plasma membrane.
- **GPI contains phosphatidylinositol**, a carbohydrate-containing linker, **and phosphate**, connecting the protein to a membrane lipid.
- GPI anchoring **helps regulate the membrane localization, stability, and function of proteins**.
- Examples of GPI-anchored proteins include alkaline phosphatase, CD55, CD59, and CD90 (Thy-1)

A GPI-anchored glycoprotein (CD90)



It is positioned in the glycolipid-rich microdomain (lipid raft) of cell plasma membrane

2026-2학기 세포의 분자적 이해

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Chapter 9.
Protein Synthesis, Processing, and Regulation (3)

분자의학교실 장병철

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